Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF MECHATRONICS ENGINEERING

VII SEMESTER B.TECH. (MECHATRONICS)

END SEMESTER EXAMINATIONS, NOVEMBER 2021

SUBJECT: PRODUCTION AND OPERATIONS MANAGEMENT [MTE 4080]

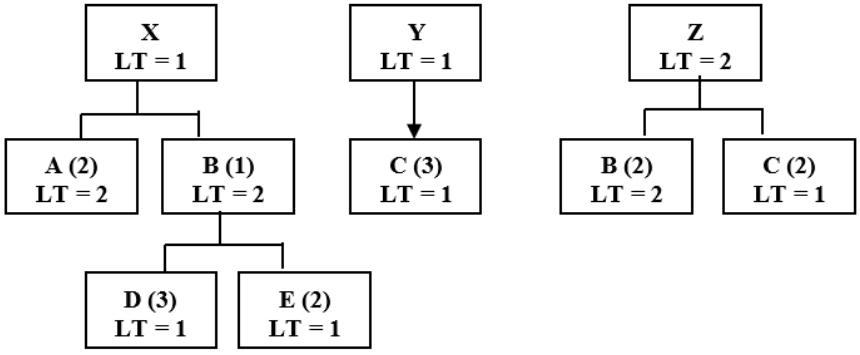
(17.11.2022)

Time: 180 MINUTES **MAX. MARKS: 50**

Instructions to Candidates:
- Answer **ALL** the questions.
- Missing data if any can be suitably assumed

Q. No	QUESTIONS	M	CO	PO	LO	BL
1A.	Draw product volume vs variety chart to represent different production systems. Cite unique characteristic and suitable example to each system.	**2**	**1**	**1, 2, 3**	**1**	**3**
1B.	A company is setting up an assembly line to produce 200 units per 10 hour shift. The following table identifies the work elements, time and immediate predecessors. Elemental tasks / Immediate predecessors / Duration of the elements (Sec.) A / --- / 30 B / A / 60 C / B / 55 D / B / 80 E / C / 85 F / D / 40 G / D / 65 H / G / 60 I / E, F, H / 35 Total time (Sec.) / 510 1. What is the desired cycle time? 2. What is the theoretical minimum number of stations? 3. Use the longest work element time rule balance the assembly line. 4. What are the resulting efficiency?	**3**	**4**	**1, 2, 3, 5**	**2**	**5**
1C.	ABC Ltd. produces three products namely X, Y and Z, which have demand, safety stock and product structure as shown in the below table and the tree structure:	**5**	**4**	**1, 2, 3, 5**	**2**	**5**

Product	Safety Stock	1	2	3	4	5	6	7	8
X	40						120		120
Y	40								200
Z	40							180	



1. On hand Inventory X = 100, Y= 100. A = 50, C = 80, E = 80.
2. The only scheduled receipts are 60 units of Z due in period 4.
3. Lot size for B is 50 and D is 50.

Determine the order quantities and order release dates for all requirements using the MRP format.

2A.	List out any four qualitative and quantitative forecasting models.	**2**	**1**	**1, 2, 3**	**1**	**3**
2B.	For a product line the annual fixed costs are Rs. 75000, the cost of production at present is Rs. 5/unit and the present sales price is Rs. 7/unit. The annual sales at present are 40000 units. Determine, a) BEP and state whether the company is having profit or loss and calculate the same. b) If the company desires to increase the BEP by 20% suggest the alternative for the same.	**3**	**2**	**1, 2, 3, 4, 5**	**3**	**4**
2C.	The following data is available for an item Annual demand = 2500units. Holding cost rate = 10% of purchase price Ordering cost = Rs. 100/order Purchase price (Rs.): 5 | 4.75 | 4.6 | 4.5 Order quantity: 100-499 | 500-2499 | 2500-4999 | 5000 and above Determine the most economical order quantity	**5**	**2**	**1, 2, 3, 4, 5**	**3**	**4**
3A	If the designed capacity of a logistics company is 50 vehicles/day. Consider 20% of vehicle capacity goes through planned maintenance everyday and if 3 vehicles delayed at maintenance, calculate utilization and efficiency.	**2**	**2**	**1, 2, 3, 4, 5**	**3**	**4**

3B For the following demand, calculate the total cost of the plan if the demand can be met by regular time production and subcontracting. | **3** | **2** | **1, 2, 3, 4, 5** | **2** | **4**

Month	Demand (Units)	Production Days
Jan	800	28
Feb	760	29
Mar	1200	28
Apr	900	26
May	1100	30

Consider regular time cost as Rs. 15 per unit and subcontracting cost as Rs. 16 per unit.

3C Two jobs are to be processed on 5 machines A,B,C,D and E. The processing time in hours and the sequence are as shown. | **5** | **3** | **1, 2, 3, 4, 5** | **3** | **5**

Job 1	A	B	C	D	E
Time	3	4	2	1	5
Job 2	C	B	D	E	A
Time	4	3	2	1	4

In what order jobs be done on each machine to minimize makespan? What is the makespan?

4A Draw the time vs quantity chart to represent manufacturing model with finite delivery without shortages. Represent all possible entities of the model. | **2** | **2** | **1, 2, 3, 4, 5** | **1** | **3**

4B Use suitable method to find the job sequencing for the following data. | **3** | **3** | **1, 2, 3, 4, 5** | **3** | **5**

Job/Machine	M1	M2	M3
Job1	15	14	14
Job2	18	16	10
Job3	16	12	14
Job4	19	18	12
Available time (Min.)	30	30	30

Also, calculate the mean flow time for the assigned sequence.

4C The monthly demand in units for ABC Ltd. For the past 6 months is as shown below. | **5** | **1** | **1, 2, 3** | **2** | **4**

Month	Jan	Feb	Mar	Apr	May	Jun
Units	100	80	110	115	105	110

Use regular straight fit line to forecast for July.

5A List out the different types of inventories present in the manufacturing units. | **2** | **4** | **1, 2, 3, 5** | **1** | **3**

5B	Discuss the production system cycle with suitable example.	**3**	**1**	**1, 2, 3**	**1**	**3**
5C	A business man decide whether to build a small capacity plant or a large capacity plant at a new location. Demand at the location can be either low or high with probabilities estimated at 0.4 and 0.6 respectively. If a small capacity plant is built and demand proves to be high the businessman may choose ether to expand or not to expand with payoffs of Rs.5,40,000 and Rs. 4,46,000 respectively. In case a small capacity plant is built and the demand is low there is no reason to expand and the payoff is Rs. 4,00,000. If a large capacity plant is built and the demand is low the choice is to do nothing with payoff of Rs. 80,000 or to stimulate demand through intensive advertising. The response to advertising may be either modest or sizable with their probabilities estimated to be 0.3 and 0.7 respectively. If it is modest the payoff is estimated to be only Rs. 40,000 and the payoff grows to Rs. 4,40,000 if the response is sizable. Finally if a large capacity plant is built and demand turns out to be high the payoff is Rs.16,00,000. Draw the decision tree and determine whether the businessman should build a small capacity plant or a large capacity plant?	**5**	**2**	**1, 2, 3, 4, 5**	**3**	**5**